Special Summer Integrated Learning Programme

Mathematics

Introduction to Differential Calculus

Week 5 Day 2 PM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

Derivatives of trigonometric functions

Theorem

- $\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\cos x = -\sin x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \sec^2 x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\csc x = -\cot x \csc x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\sec x = \sec x \tan x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\cot x = -\csc^2 x.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\arcsin x = \frac{1}{\sqrt{1-x^2}}.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\arccos x = \frac{-1}{\sqrt{1-x^2}}.$
- $\frac{\mathrm{d}}{\mathrm{d}x}\arctan x = \frac{1}{1+x^2}.$

The derivative of sine can be derived rigorously by the addition formula and the squeeze theorem, which we'll omit here. It can also be explained intuitively using geometry, here's a video that does that. All derivatives of the other trigonometric functions can be derived from the derivative of sine.
As for the derivatives of inverse trigonometric functions, they can be derived from implicit differentiation, which you'll learn in Calculus in term 1.

Derivatives of trigonometric functions - proof

Given $\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x$, then by the cofunction identity and the chain rule, we derive that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\cos x &= \frac{\mathrm{d}}{\mathrm{d}x}\sin\left(\frac{\pi}{2} - x\right)\\ &= \cos\left(\frac{\pi}{2} - x\right)\cdot(-1)\\ &= -\sin x.\end{aligned}$$

On the other hand, by the quotient rule, we derive that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\tan x &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin x}{\cos x}\\ &= \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)}\\ &= \frac{1}{\cos^2(x)} = \sec^2(x).\end{aligned}$$

Activity 1

(a) Derive the derivatives of $\csc x$, $\sec x$ and $\cot x$ using trigonometric identities and the chain rule.

(b) Find the derivatives of the following functions.

(i) $\tan(x^2 + x)$

(ii) $\arctan\left(\dfrac{x^2+1}{x^2-1}\right)$

(iii) $\dfrac{\csc x + \sec x}{x+1}$

(iv) $\sec(\tan x)$

(v) $\arcsin(\sqrt{x})$

Activity 1: solution

(a) Since $\csc x = (\sin x)^{-1}$, so

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\csc x &= \frac{\mathrm{d}}{\mathrm{d}x}(\sin x)^{-1}\\ &= -(\sin x)^{-2}\cdot\cos x\\ &= -\frac{\cos x}{\sin x}\frac{1}{\sin x}\\ &= -\cot x\csc x.\end{aligned}$$

Since $\sec x = (\cos x)^{-1}$, so

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\sec x &= \frac{\mathrm{d}}{\mathrm{d}x}(\cos x)^{-1}\\ &= -(\cos x)^{-2}\cdot(-\sin x)\\ &= \frac{1}{\cos x}\frac{\sin x}{\cos x}\\ &= \sec x\tan x.\end{aligned}$$

Activity 1: solution

(a) Since $\cot x = (\tan x)^{-1}$, so

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\cot x &= \frac{\mathrm{d}}{\mathrm{d}x}(\tan x)^{-1}\\ &= -(\tan x)^{-2}\cdot\sec^2 x\\ &= -\frac{\cos^2 x}{\sin^2 x}\frac{1}{\cos^2 x}\\ &= -\csc^2 x.\end{aligned}$$

(b) (i)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\tan(x^2+x) &= \sec^2(x^2+x)\cdot(2x+1)\\ &= (2x+1)\sec^2(x^2+x).\end{aligned}$$

Activity 1: solution

(b) (ii)

$$
\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}x}\arctan\left(\frac{x^2+1}{x^2-1}\right)\\
&=\frac{1}{1+\left(\frac{x^2+1}{x^2-1}\right)^2}\cdot\frac{(2x)(x^2-1)-(2x)(x^2+1)}{(x^2-1)^2}\\
&=\frac{(x^2-1)^2}{(x^2-1)^2+(x^2+1)^2}\cdot\frac{-4x}{(x^2-1)^2}\\
&=\frac{-4x}{(x^2-1)^2+(x^2+1)^2}\\
&=\frac{-4x}{x^4-4x^2+1+x^4+4x^2+1}\\
&=\frac{-4x}{2x^4+2}=\frac{-2x}{x^4+1}.
\end{aligned}
$$

Activity 1: solution

(b) (iii)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\csc x+\sec x}{x+1}\right) &= \frac{(-\csc x\cot x+\sec x\tan x)(x+1)-(\csc x+\sec x)\cdot 1}{(x+1)^2}\\ &= \frac{((x+1)\tan x-1)\sec x-((1+x)\cot x+1)\csc x}{(x+1)^2}.\end{aligned}$$

(iv)

$$\frac{\mathrm{d}}{\mathrm{d}x}\sec(\tan x)=\sec(\tan x)\tan(\tan x)\cdot\sec^2 x.$$

(v)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\arcsin(\sqrt{x}) &= \frac{1}{\sqrt{1-(\sqrt{x})^2}}\cdot\frac{1}{2\sqrt{x}}\\ &= \frac{1}{2\sqrt{x}\sqrt{1-x}}.\end{aligned}$$

Outline

Derivatives of exponential and logarithmic functions

Theorem

For any constant $a > 0$, we have

- $$\frac{\mathrm{d}}{\mathrm{d}x} a^x = a^x \ln a,$$
  in particular
  $$\frac{\mathrm{d}}{\mathrm{d}x} e^x = e^x.$$
- $$\frac{\mathrm{d}}{\mathrm{d}x} \log_a(x) = \frac{1}{x \ln a,}$$
  in particular
  $$\frac{\mathrm{d}}{\mathrm{d}x} \ln x = \frac{1}{x}.$$

We omit the proofs here, which you'll see in Calculus in term 1.

Derivatives of exponential and logarithmic functions - example

E.g.: $\frac{\mathrm{d}}{\mathrm{d}x}\left(2 \cdot 3^x + 5e^x\right) = 2 \ln 3 \cdot 3^x + 5e^x.$

E.g.: $\frac{\mathrm{d}}{\mathrm{d}x} e^x \ln x = e^x \ln x + \frac{e^x}{x}.$

E.g.: $\frac{\mathrm{d}}{\mathrm{d}x} x \ln x = 1 \cdot \ln x + \frac{x}{x} = \ln x + 1.$

E.g.: $\frac{\mathrm{d}}{\mathrm{d}x} \ln(\cos x) = \frac{1}{\cos x} \cdot (-\sin x) = -\tan x.$

E.g.: $\frac{\mathrm{d}}{\mathrm{d}x} \lg(x^3 + 1) = \frac{1}{(x^3+1)\ln 10} \cdot (3x^2) = \frac{3x^2}{(x^3+1)\ln 10}.$

Activity 2

Differentiate the following functions with respect to x.

(a) $2e^x + x^2$

(b) e^{x^2}

(c) $\sqrt{1 + e^{\tan x}}$

(d) $\ln\left(\dfrac{1+x}{1-x}\right)$

(e) $(x^2 + 3x)(1 - e^{-2x})$

(f) $\sec(\pi^x)$

(g) $\arctan(e^x + e^{-x})$

(h) $\log_2(\sqrt{x^4 + 1})$

Activity 2: solution

(a)

$$\frac{\mathrm{d}}{\mathrm{d}x}(2e^x + x^2) = 2e^x + 2x.$$

(b)

$$\frac{\mathrm{d}}{\mathrm{d}x}e^{x^2} = e^{x^2} \cdot (2x) = 2xe^{x^2}.$$

(c)

$$\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{1 + e^{\tan x}} = \frac{1}{2\sqrt{1 + e^{\tan x}}} \cdot e^{\tan x} \cdot \sec^2 x = \frac{e^{\tan x}\sec^2 x}{2\sqrt{1 + e^{\tan x}}}.$$

Activity 2: solution

(d)

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{1+x}{1-x}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(1+x)-\ln(1-x)\right)$$
$$= \frac{1}{1+x} - \frac{1}{1-x}(-1)$$
$$= \frac{1}{1+x} + \frac{1}{1-x} = \frac{2}{1-x^2}.$$

(e)

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2+3x)(1-e^{-2x})$$
$$= (2x+3)(1-e^{-2x}) + (x^2+3x)(-e^{-2x}(-2))$$
$$= 2x+3+e^{-2x}(2x^2+4x-3).$$

(f)

$$\frac{\mathrm{d}}{\mathrm{d}x}\sec(\pi^x) = \sec(\pi^x)\tan(\pi^x)\pi^x\ln\pi.$$

Activity 2: solution

(g)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\arctan(e^x+e^{-x}) &= \frac{1}{1+(e^x+e^{-x})^2}\cdot(e^x-e^{-x})\\ &= \frac{e^x-e^{-x}}{1+e^{2x}+2+e^{-2x}}\\ &= \frac{e^x-e^{-x}}{3+e^{2x}+e^{-2x}}\end{aligned}$$

(h)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\log_2(\sqrt{x^4+1}) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\log_2(x^4+1)\right)\\ &= \frac{1}{2}\frac{1}{(x^4+1)\ln 2}\cdot(4x^3)\\ &= \frac{2x^3}{(x^4+1)\ln 2}.\end{aligned}$$

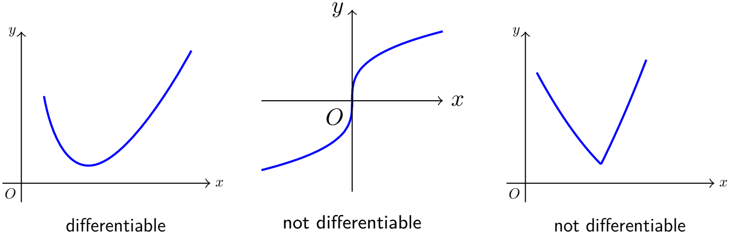
Outline

1. Derivatives of Trigonometric Functions
2. Derivatives of Exponential and Logarithmic Functions
3. Higher Order Derivatives
4. Summary

Differentiability

Definition:

A function f is ***differentiable on an interval*** (a, b) if the derivative exists at every point of (a, b). In general, if the derivative exists at every point in the domain of f, then we say that f is ***differentiable***.



Differentiability implies continuity but not the converse. For instance, $f(x) = |x|$ is continuous at $x = 0$, but not differentiable there.

Differentiable functions are continuous

Theorem

If $f(x)$ is differentiable at x_0, then $f(x)$ is continuous at x_0.

Proof:

$$\begin{aligned}\lim_{x\to x_0}(f(x)-f(x_0)) &= \lim_{x\to x_0}\left(\frac{f(x)-f(x_0)}{x-x_0}(x-x_0)\right)\\ &= \lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}\lim_{x\to x_0}(x-x_0)\\ &= f'(x_0)\times 0 = 0.\end{aligned}$$

Therefore

$$\begin{aligned}\lim_{x\to x_0} f(x) &= \lim_{x\to x_0}(f(x)-f(x_0)+f(x_0))\\ &= \lim_{x\to x_0}(f(x)-f(x_0)) + \lim_{x\to x_0} f(x_0) = 0 + f(x_0) = f(x_0).\end{aligned}$$

So $f(x)$ is continuous at x_0. Q.E.D.

n-th order derivatives

Since $f'(x)$ is itself a function, it's possible to ask whether f' is differentiable. This is called the 2nd order derivative of $f(x)$.

Definition:

The ***2nd order derivative*** of $f(x)$, denoted as $f''(x)$ or $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}(x)$, is the derivative of $f'(x)$, i.e.

$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}(x) = f''(x) := \frac{\mathrm{d}f'}{\mathrm{d}x}(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}.$$

Similarly, the ***3rd order derivative*** of $f(x)$ is $\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}(x) = f'''(x) := \frac{\mathrm{d}f''}{\mathrm{d}x}(x)$. The ***$n$-th order derivative*** of $f(x)$, denoted as $\frac{\mathrm{d}^n f}{\mathrm{d}x^n}$, is the function obtained by successively differentiating $f(x)$ n times, if it exists.

E.g.: For $f(x) = \sin(2x)$, then $f'(x) = 2\cos(2x)$, $f''(x) = -4\sin(2x)$, $f'''(x) = -8\cos(2x)$, $f''''(x) = 16\sin(2x)$.

n-TH ORDER DERIVATIVES

Notation: We also denote the n-th order derivative of $f(x)$ by $f^{(n)}(x)$. This notation is preferred when the order of derivative is high. Between $f''''''''(x)$ and $f^{(8)}(x)$, which one can you tell the order of derivative immediately?

E.g.: For $f(x) = x^3$, $f^{(1)}(x) = 3x^2$, $f^{(2)}(x) = 6x$, $f^{(3)}(x) = 6$ and $f^{(n)}(x) = 0$ for all integers $n \geq 4$.

⚠ Do not confuse $f^{(n)}(x)$ with $f^n(x) := (f(x))^n$.

E.g.: For $f(x) = x^3$, $f^2(x) = (f(x))^2 = (x^3)^2 = x^3 \cdot x^3 = x^6$.

Second order derivative $f''(x)$

Since

$$f''(x) = \frac{\mathrm{d}}{\mathrm{d}x} f'(x),$$

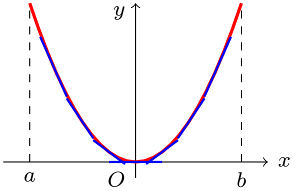
the second derivative is the rate of change of $f'(x)$ w.r.t. x, i.e. the rate of change of the slope of the tangent to the graph of $y = f(x)$ w.r.t. x.

If $f''(x) \geq 0$ over an interval I, then its slope is non-decreasing, i.e. the graph of $f(x)$ resembles (part of) a $\bigcup$ shape over I.
If $f''(x) \leq 0$ over an interval I, then its slope is non-increasing, i.e. the graph of $f(x)$ resembles (part of) a $\bigcap$ shape over I.

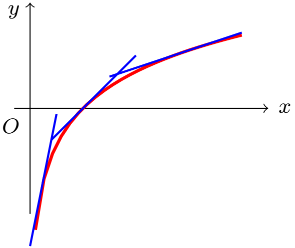
Definition:

If $f''(x) \geq 0$ over an interval I, then f is ***convex*** over I. If $f''(x) \leq 0$ over an interval I, then f is ***concave*** over I. An ***inflection point*** is a point at which f changes its convexity/concavity.

Convex and concave functions



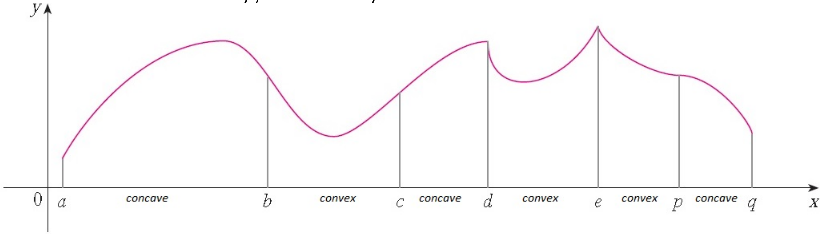
Geometrically, a function is convex if its graph lies above or coincides with all of its tangent lines.



Geometrically, a function is concave if its graph lies below or coincides with all of its tangent lines.

Convex and concave functions

In general, a function $f(x)$ is convex on some intervals and concave on other intervals. By knowing the intervals on which $f''(x) \geq 0$ and $f''(x) \leq 0$, we can sketch the graph of $f(x)$ with the correct convexity/concavity.

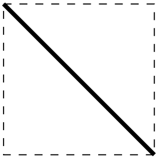


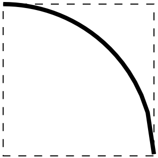
The inflection points occur at $x = b, c, d$ and p, because there are changes in convexity/concavity there.

⚠ $f''(x_0) = 0$ doesn't imply that there's an inflection point at $x = x_0$. For instance, $f(x) = x^4$ is convex over $\mathbb{R}$ and hence it has no inflection point, though $f''(0) = 0$.

Activity 3

Use a thick, black marker to draw on square sticky notes to create 'linear' and 'curvy' puzzle pieces as shown below. Make 4 of each.





(a) Use the provided information on each line to fill in the remaining columns. A couple of examples are provided for you.
Graph A

Interval	Description in words	Description in symbols	Sketch
1	convex & increasing		
2	concave & increasing	$f''(x) \leq 0$ & $f'(x) > 0$	
3	zero convexity & decreasing		
4	zero slope		
5	zero convexity & increasing	$f''(x) = 0$ & $f'(x) > 0$	

Activity 3

(a) Graph B

Interval	Description in words	Description in symbols	Sketch
1		$f''(x) = 0 \ \& \ f'(x) > 0$	
2		$f''(x) = 0 \ \& \ f'(x) < 0$	
3		$f''(x) \geq 0 \ \& \ f'(x) < 0$	
4	concave & decreasing	$f''(x) \leq 0 \ \& \ f'(x) < 0$	
5		$f'(x) = 0$	

Graph C

Interval	Description in words	Description in symbols	Sketch
1	concave & decreasing		
2			
3		$f'(x) = 0$	
4		$f''(x) = 0 \ \& \ f'(x) > 0$	
5	zero convexity & decreasing		

(b) Now that you've completed the tables, assemble the graphs of each of the continuous functions described.

Activity 3: solution

(a) Graph A

Interval	Description in words	Description in symbols	Sketch
1	convex & increasing	$f''(x) \geq 0 \;\; \& \;\; f'(x) > 0$	
2	concave & increasing	$f''(x) \leq 0 \;\; \& \;\; f'(x) > 0$	
3	zero convexity & decreasing	$f''(x) = 0 \;\; \& \;\; f'(x) < 0$	
4	zero slope	$f'(x) = 0$	
5	zero convexity & increasing	$f''(x) = 0 \;\; \& \;\; f'(x) > 0$	

Graph B

Interval	Description in words	Description in symbols	Sketch
1	zero convexity & increasing	$f''(x) = 0 \;\; \& \;\; f'(x) > 0$	
2	zero convexity & decreasing	$f''(x) = 0 \;\; \& \;\; f'(x) < 0$	
3	convex & decreasing	$f''(x) \geq 0 \;\; \& \;\; f'(x) < 0$	
4	concave & decreasing	$f''(x) \leq 0 \;\; \& \;\; f'(x) < 0$	
5	zero slope	$f'(x) = 0$	

Graph C

Interval	Description in words	Description in symbols	Sketch
1	concave & decreasing	$f''(x) \leq 0 \;\; \& \;\; f'(x) < 0$	
2	convex & decreasing	$f''(x) \geq 0 \;\; \& \;\; f'(x) < 0$	
3	zero slope	$f'(x) = 0$	
4	zero convexity & increasing	$f''(x) = 0 \;\; \& \;\; f'(x) > 0$	
5	zero convexity & decreasing	$f''(x) = 0 \;\; \& \;\; f'(x) < 0$	

Activity 4

Given $f(x) = x^4(x^2 - 1)$, find the interval over which f is convex and the interval over which f is concave. Hence, find the inflection point(s) of f.

Activity 4: solution

$f(x) = x^4(x^2 - 1) = x^6 - x^4$. So

$$f'(x) = 6x^5 - 4x^3$$

and

$$f''(x) = 30x^4 - 12x^2 = 6x^2(5x^2 - 2).$$

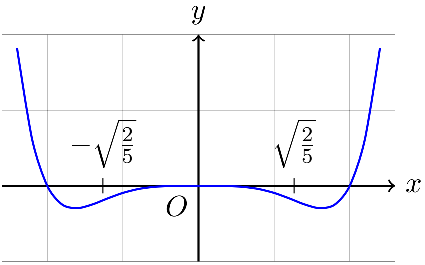
Thus,

$$\begin{aligned} f''(x) \geq 0 \quad &\Leftrightarrow \quad 6x^2(5x^2 - 2) \geq 0 \\ &\Leftrightarrow \quad 5x^2 - 2 \geq 0 \quad \text{or} \quad x = 0 \\ &\Leftrightarrow \quad x^2 \geq \frac{2}{5} \quad \text{or} \quad x = 0 \\ &\Leftrightarrow \quad x \leq -\sqrt{\frac{2}{5}} \quad \text{or} \quad x \geq \sqrt{\frac{2}{5}} \quad \text{or} \quad x = 0. \end{aligned}$$

Activity 4: solution

Since f is convex iff. $f''(x) \geq 0$, so f is convex over the intervals $\left(-\infty, -\sqrt{\frac{2}{5}}\right]$ and $\left[\sqrt{\frac{2}{5}}, \infty\right)$. On the other hand, f is concave iff. $f''(x) \leq 0$, so f is concave over the interval $\left[-\sqrt{\frac{2}{5}}, \sqrt{\frac{2}{5}}\right]$.

Since there's a change in convexity/concavity at $x = -\sqrt{\frac{2}{5}}$ and $x = \sqrt{\frac{2}{5}}$, these are all the inflection points of f.



Outline

Summary

- Derivatives of Trigonometric Functions
  - derivatives of sine, cosine, tangent, secant, cosecant and cotangent.
  - derivatives of arcsine, arccosine and arctangent.
- Derivatives of Exponential and Logarithmic Functions
  - derivatives of a^x and $\log_a(x)$.
- Higher Order Derivatives
  - differentiable functions.
  - differentiability implies continuity.
  - n-th order derivatives.
  - the second order derivative, convex and concave functions, inflection points.

Resources

References:

- George F. Simmons. Calculus with Analytic Geometry (2nd edition) sections 3.4, 8.3, 8.4, 4.2.
- Hughes-Hallett. Calculus (6th edition) sections 3.5, 3.2, 2.5.